

TED UNIVERSITY

CMPE 491 – Senior Project I

Fall 2026 – Vulnerability Prioritization Assistant

Project Proposal

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Project Description

Security scanners return hundreds or thousands of findings even for a small environment, but teams can fix only a few each week, and a CVSS score alone does not show which ones matter. A 9.8 on an unimportant test machine can outrank a 7.5 on an internet-facing web server that attackers are actively exploiting. The Vulnerability Prioritization Assistant turns raw scanner output into a short, ordered to-do list in which every item has a plain-language reason and a suggested fix.

The system converts vulnerability scanner output into one common format and enriches each finding with free public data: CISA KEV, EPSS and NVD. It then computes a transparent, tunable priority score from severity, evidence of exploitation and asset context such as internet exposure and data sensitivity. We will measure the resulting ranking against a baseline that uses CVSS alone.